
Breaking the Curse of Repulsion: Distributionally Robust Policy Optimization for Off-Policy Learning

Jie Jiang^{*1} Yusen Huo^{*1} Xiangxin Zhan¹ Changping Wang¹ Jun Zhang¹

Abstract

Negative feedback is a central resource in policy optimization: it suppresses known bad behavior and can move a policy beyond the solution obtained from positive examples alone. Yet the same feedback can become destructive when historical negative actions remain in offline data, replay, or stale trajectories after the learner has moved away from them. We identify the missing mechanism by separating sample badness from learner-relative distance or rarity, showing that policy remoteness independently amplifies negative influence. We then develop Repulsive Dynamics, which characterizes a transition from the Positive-only target, through stable extrapolation, to a feasibility boundary and the loss of a finite equilibrium. Based on this analysis, Distributionally Robust Policy Optimization (DRPO) reweights the aggregate negative term with a policy-relative exponential envelope, retaining useful local repulsion while suppressing its far-field tail. Controlled continuous and categorical environments isolate the source and causal transmission of the effect; registered Hopper/D4RL and Countdown experiments are reserved for external validation under learned critics and shared-parameter sequence policies, with formal results remaining TBD until terminal audit. The resulting framework tests a general principle: policy optimization should control far-field negative influence rather than remove negative feedback as a whole.

1. Introduction

Off-policy policy optimization improves a current policy by reusing behavior collected by earlier, stale, or heterogeneous policies. This paradigm is central to offline reinforcement learning, replay-based control, recommender systems, and model post-training because it reduces the need for fresh

interaction and exploits experience accumulated across policy iterations (Levine et al., 2020; Fujimoto et al., 2019). Historical data contain two complementary signals. Positive updates increase the likelihood of observed successful actions, whereas negative feedback suppresses known failures and competing modes (Peng et al., 2019; Wang et al., 2020). The Positive-only endpoint, which removes every negative update, therefore yields a useful stability reference but can restrict the learner to attraction toward the empirical positive target. Balanced repulsion can instead reshape the decision boundary and move the aggregate equilibrium beyond that target. The central question is not whether negative feedback should be kept or discarded wholesale, but how to preserve its policy-improvement value without allowing repeated reuse to turn it into excessive repulsion.

The difficulty appears when negative actions are historical rather than freshly sampled. Offline logs, replay buffers, stale actors, and asynchronous trajectories can keep presenting the same action after the current learner has changed (Schulman et al., 2017; Haarnoja et al., 2018). A negative sample may initially lie near the policy and provide locally informative boundary information. Its update then pushes the learner away, increasing the action’s learner-relative distance or surprisal while the stored negative label remains active. In Gaussian policies, the mean score can grow with standardized distance, so subsequent reuse produces progressively larger repulsion. In categorical policies, the direct-logit score is bounded, yet repeated negative updates continue to lower the selected action’s log-odds and can drive its probability toward the support boundary. Historical reuse therefore creates a policy-family-dependent transition from useful local feedback to destructive far-field influence, rather than merely exposing the learner to one unusually large update.

Existing methods already regulate negative or off-policy updates in several principled ways. Positive-only objectives remove negative terms; global coefficients and clipping reduce their scale; trust regions and behavior constraints restrict policy movement; rarity-aware rules target low-probability events; and quality filtering removes selected data (Schulman et al., 2017; Kumar et al., 2020; Kostrikov et al., 2021; Peng et al., 2019). These controls can improve stability, but

^{*}Equal contribution ¹Tencent Inc, China. Correspondence to: Jun Zhang <neoxzhang@tencent.com>.

they do not by themselves explain why the same negative action can be useful near the learner and destructive in the far field after repeated reuse. The issue is difficult to identify in ordinary logs because low reward, large negative advantage, rarity, and learner-relative distance are typically correlated. We therefore construct matched controls that use matching context and semantic role, holding negative-advantage magnitude, sample count, base coefficient, and policy stage fixed to isolate policy remoteness. The source-identification protocol tests whether remoteness independently changes gradient scale; a separate intervention protocol tests whether the resulting far-field influence transmits into drift, support-boundary events, and task-performance collapse. This separation motivates an aggregate theory rather than another static weighting heuristic.

Repulsive Dynamics separates three layers that are often conflated. First, learner-relative remoteness determines the score response of a fixed historical action. Second, repeated reuse creates an asymmetry: positive updates make a successful action more compatible with the learner and attenuate its score, whereas negative updates make a rejected action more remote and preserve or amplify its response. Third, positive and negative contributions aggregate into a signed policy field. When positive mass dominates and the signed target remains inside the feasible mean-parameter space, the policy has a finite stable equilibrium beyond the Positive-only target. At the critical balance the field can produce persistent drift, while negative dominance makes any finite stationary point unstable. The terminal manifestation depends on the policy family: Gaussian mean runaway yields unbounded far-field scores, whereas categorical policies approach the simplex boundary with bounded per-sample logit scores. Negative feedback is therefore neither uniformly beneficial nor uniformly harmful; its value depends on aggregate balance and learner-relative location.

DRPO acts on the mechanism identified by the theory rather than treating every negative sample as toxic. Starting from the signed empirical actor field, it reweights the aggregate negative term by learner-relative distance or surprisal. Nearby negative actions retain substantial weight and can continue to shape local boundaries or suppress competing bad modes. As an action becomes increasingly remote, the weight decays nonlinearly, preventing the far-field tail from dominating the positive attraction. The exponential envelope is chosen for a precise tail property: under finite-order growth of the unweighted score-times-advantage contribution, the weighted contribution vanishes as remoteness increases. This guarantee does not assume that a sample’s utility decays exponentially and does not establish a universal ranking over all tapers. Quality-based selection and learner-relative remoteness control are distinct axes; the experiments evaluate them and global scaling under matched conditions rather than presuming that one control must al-

ways win.

The empirical program follows the same decomposition as the theory. First, we ask whether far-field or rare negative updates become disproportionately influential in realistic policy learning. Second, using matched quality–distance and quality–rarity controls, we test whether remoteness independently changes negative influence and whether targeted far-field interventions interrupt drift or boundary events. Third, we test the predicted sequence from the Positive-only platform through stable extrapolation to persistent drift or instability, and compare selective attenuation with global and budget-matched controls. Fourth, we evaluate whether the resulting control improves external tasks. C-U1 and D-U1 provide controlled continuous and categorical identification; Hopper/D4RL and Countdown provide external validity and do not replace those controlled mechanisms. Every dynamics claim is terminal-audited, and task-performance collapse, support or variance-boundary events, and NaN/Inf numerical failure are reported separately. We contribute a layered theory of repeated repulsion, matched causal identification of learner-relative remoteness, DRPO control of the far-field negative term, and an evidence chain that keeps mechanism claims distinct from external task claims.

2. Related Work

Quality-aware policy fitting already provides several ways to use or suppress suboptimal behavior. Advantage-weighted regression, implicit value learning, critic-regularized regression, and proximal behavior objectives emphasize actions estimated to be better while reducing the impact of lower-quality data (Peng et al., 2019; Kostrikov et al., 2021; Wang et al., 2020; Zhuang et al., 2023). Positive-only and hard-filtering variants take the conservative endpoint and remove selected negative updates altogether. At the same time, failed or suboptimal behavior can remain informative: it can suppress competing bad modes, sharpen a decision boundary, or release probability mass for alternatives represented by the model. The established picture is therefore not that negative feedback is simply noise, but that it combines information value with optimization risk. What remains under-characterized is the temporal relation to the current learner. A fixed negative action can begin as relevant local feedback and become increasingly remote after repeated reuse. We study that transition and its aggregate consequences, complementing rather than dismissing prior weighting and filtering methods.

Off-policy learning has long controlled mismatch between the behavior distribution and the current policy. Importance correction, PPO-style clipping, and trust regions limit the effect of samples collected by another policy (Schulman et al., 2017). Replay-based algorithms and maximum-entropy off-policy control likewise manage repeated data reuse while the

learner evolves (Haarnoja et al., 2018). Offline-RL analyses make the same mismatch explicit when no new interaction is available (Levine et al., 2020; Fu et al., 2020). Low action probability or high surprisal can therefore serve as a useful learner-relative warning signal. These approaches establish that mismatch and rarity matter; our distinction concerns how they arise. We treat remoteness not only as a static property used to choose a weight, but as an endogenous state variable that a negative update increases and subsequent reuse feeds back into the next update. We then connect that per-sample process to aggregate equilibria and targeted causal interventions. This perspective complements clipping and stale-policy control rather than asserting that they are ineffective.

Offline reinforcement learning contains several distinct failure channels, and far-field signed actor dynamics is only one of them. Conservative Q-learning controls value extrapolation by assigning pessimistic values to unsupported actions (Kumar et al., 2020). Implicit Q-learning avoids querying out-of-distribution actions during value improvement and then fits an actor to supported high-value behavior (Kostrikov et al., 2021). Behavior-constrained and proximal approaches instead keep the learned policy close to the data or a behavior reference (Fujimoto et al., 2019; Wang et al., 2020; Zhuang et al., 2023). Data selection adds another axis by changing which transitions participate in learning. DRPO focuses on the signed actor field after an advantage-like signal has been supplied: it asks how negative actor updates change as historical actions become remote, and how to preserve their local information without allowing their far-field contribution to dominate. This actor-side control can be combined with pessimistic critics, in-support value learning, or behavior regularization. It is not presented as a replacement for those safeguards or as a complete explanation of every offline-RL failure.

3. Problem Setup

We analyze the actor-side dynamics generated by repeatedly reusing signed feedback from a historical update distribution. Let ν denote a distribution over state–action pairs, representing an offline dataset, a replay buffer, or trajectories collected by earlier or stale policies, and let $\hat{A}(s, a)$ be the advantage-like coefficient supplied to the actor. During one actor step, both ν and $\hat{A}$ are treated as fixed with respect to θ . The resulting empirical actor field and discrete update are

$$\begin{aligned} \mathbf{F}(\theta) &= \mathbb{E}_{(s,a) \sim \nu} \left[\hat{A}(s, a) \nabla_{\theta} \log \pi_{\theta}(a | s) \right], \\ \theta_{t+1} &= \theta_t + \eta \mathbf{F}(\theta_t). \end{aligned} \quad (1)$$

This field is not assumed to equal the exact on-policy policy gradient, because ν need not match the current state–action distribution (Sutton & Barto, 2018; Levine et al., 2020).

Define the positive and negative components

$$\hat{A}^+ = \max(\hat{A}, 0), \quad \hat{A}^- = \max(-\hat{A}, 0), \quad (2)$$

so that $\mathbf{F}$ decomposes into attraction from $\hat{A}^+$ and repulsion from $\hat{A}^-$. For one negative sample, the update magnitude factorizes as

$$\left\| \hat{A}^- \nabla_{\theta} \log \pi_{\theta}(a | s) \right\|_2 = \hat{A}^- \left\| \nabla_{\theta} \log \pi_{\theta}(a | s) \right\|_2. \quad (3)$$

The first factor measures disadvantage severity; the second is learner-relative score geometry. At the aggregate level, sample frequency and directional coherence further determine whether these contributions cancel or reinforce. Freezing ν and $\hat{A}$ is therefore a mechanism-isolation device: it asks what the actor score alone can do under historical reuse, without claiming that critic error, importance ratios, or jointly evolving actor–critic dynamics are absent in full algorithms.

To compare continuous and categorical policies without forcing them to share the same amplification law, we use a common learner-relative coordinate. For a fixed context s and historical action a , define the negative log probability

$$D_{\theta}(s, a) = -\log \pi_{\theta}(a | s). \quad (4)$$

For a Gaussian policy, D_{θ} equals a scale-dependent constant plus one half of the squared Mahalanobis distance from the policy mean. For a categorical policy, it is exactly the selected action’s surprisal. In sequence models the same quantity may be evaluated at token level or normalized across a completion, but the normalization must be fixed before comparing examples. Crucially, remoteness is dynamic: the historical action and its label remain fixed while $D_{\theta}(s, a)$ changes as the learner moves. We separately define the squared score response

$$R_{\theta}(s, a) = \left\| \nabla_{\theta} \log \pi_{\theta}(a | s) \right\|_2^2. \quad (5)$$

The notation is shared, but no universal relation between D_{θ} and R_{θ} is assumed. Gaussian mean coordinates can have unbounded response, categorical direct-logit scores are bounded, and shared neural parameters introduce an additional Jacobian. Proposition 4.1 therefore derives the policy-family-specific static relation before we analyze repeated reuse.

4. Repulsive Dynamics

We begin with a static relation: at a fixed learner, remoteness controls the score response of a historical action. Fix a context and write u for the policy-output coordinates. Recall from Equations (4)–(5) that $D_u(a) = -\log \pi_u(a)$ and $R_u(a) = \left\| \nabla_u \log \pi_u(a) \right\|_2^2$.

Proposition 4.1 (Learner-relative remoteness and score response). *For $\pi_\mu = \mathcal{N}(\mu, \Sigma)$ with fixed $\Sigma \succ 0$ and $C_\Sigma = \frac{d}{2} \log(2\pi) + \frac{1}{2} \log |\Sigma|$,*

$$\begin{aligned} R_\mu(a) &\geq 2\lambda_{\min}(\Sigma^{-1})(D_\mu - C_\Sigma), \\ R_\mu(a) &\leq 2\lambda_{\max}(\Sigma^{-1})(D_\mu - C_\Sigma). \end{aligned} \quad (6)$$

Along any fixed direction the relation is exactly linear in $D_\mu - C_\Sigma$; in the isotropic case $\Sigma = \sigma^2 I$, it reduces to $R_\mu(a) = 2(D_\mu - C_\Sigma)/\sigma^2$. For a categorical policy with selected-action probability $p_a = e^{-D_z(a)}$,

$$|\partial_{z_a} \log \pi_z(a)| = 1 - e^{-D_z(a)}, \quad \|\nabla_z \log \pi_z(a)\|_2^2 \leq 2. \quad (7)$$

The Gaussian mean score is therefore unbounded as Mahalanobis remoteness increases, with the covariance eigenvalues controlling the amplification rate. Under isotropic covariance, remoteness and Gaussian mean-score magnitude have the same global ordering; under anisotropic covariance the ordering remains bounded by the two eigenvalue slopes. Categorical rarity also strengthens selected-logit suppression, but the response saturates: low probability does not imply an unbounded direct-logit gradient. This distinction matters because a categorical policy may still undergo persistent support suppression even though every per-sample score is finite. Proposition 4.1 is only a comparison at one fixed learner. It does not establish aggregate divergence, task-performance collapse, or NaN/Inf failure; Appendix A.1 gives the complete derivation. We next study how repeated reuse changes both remoteness and score response over time.

Historical reuse converts the static score relation into opposite positive and negative feedback loops. Fix a historical action a , write $D(u) = -\log \pi_u(a)$ and $R(u) = \|\nabla_u D(u)\|_2^2$, and consider the repeated update

$$u_{t+1} = u_t - \eta \hat{A} \nabla_u D(u_t). \quad (8)$$

The action and its signed coefficient remain fixed over the reuse window; only the learner coordinates change.

Theorem 4.2 (Self-attenuation and self-amplification under reuse). *Assume that D is differentiable and convex along the update path. If $\hat{A} < 0$, then every negative reuse step satisfies*

$$D_{t+1} \geq D_t + \eta |\hat{A}| R_t, \quad R_{t+1} \geq R_t. \quad (9)$$

If $\hat{A} > 0$, ∇D is L -Lipschitz, and $0 < \eta \hat{A} \leq 1/L$, then positive reuse satisfies

$$D_{t+1} \leq D_t - \frac{\eta \hat{A}}{2} R_t, \quad R_{t+1} \leq R_t. \quad (10)$$

The first pair of inequalities says that negative reuse makes the fixed historical action more remote and cannot weaken

its subsequent score response. The second pair gives the complementary positive mechanism: sufficiently small positive reuse makes the successful action more compatible with the learner and attenuates its response. Convexity holds for fixed-covariance Gaussian mean coordinates and categorical logits, and for regular exponential families in natural coordinates because $\nabla^2 D$ is a covariance matrix. Combining this theorem with Proposition 4.1 yields unbounded Gaussian mean-score amplification but bounded, persistent categorical suppression. The result does not determine the aggregate policy trajectory: positive samples regain attractive force as the learner moves, and updates from different samples may cancel. Appendix A.2 proves the inequalities and the score monotonicity. We therefore turn from the single-sample loop to the aggregate balance of positive and negative historical updates.

Per-sample amplification does not imply global divergence because the full field balances all positive and negative historical updates. At a fixed context, consider a regular minimal exponential-family policy $\pi_\eta(a) = h(a) \exp\{\eta^\top T(a) - \psi(\eta)\}$. Let $p = \mathbb{E}_\nu[\hat{A}^+]$ and $q = \mathbb{E}_\nu[\hat{A}^-]$ denote the effective positive and negative masses, and let m_+ and m_- be their normalized sufficient-statistic moments. Expanding the signed objective gives the aggregate field

$$\mathbf{F}(\eta) = pm_+ - qm_- - (p - q)\nabla\psi(\eta). \quad (11)$$

This expression separates the fixed signed target $pm_+ - qm_-$ from the policy-dependent mean parameter $\nabla\psi(\eta)$. It is the exact population field for the frozen signed-update model.

Theorem 4.3 (Aggregate attraction–repulsion equilibria). *If $p > q$, define*

$$m^* = \frac{pm_+ - qm_-}{p - q}. \quad (12)$$

When m^ lies in the interior mean-parameter space, there is a unique finite equilibrium η^* satisfying $\nabla\psi(\eta^*) = m^*$. It is locally asymptotically stable in continuous time and locally stable for a sufficiently small discrete step size. Moreover,*

$$m^* - m_+ = \frac{q}{p - q}(m_+ - m_-), \quad (13)$$

so negative feedback moves the equilibrium beyond the Positive-only target in the direction away from the negative moment. As m^ approaches the feasible boundary, the realizing natural parameter leaves every compact set; outside the feasible set no finite equilibrium exists. If $p = q$ and $m_+ \neq m_-$, the field is constant and produces persistent drift. If $p < q$, any finite stationary point is unstable in continuous and discrete time.*

The positive-dominant regime therefore permits finite stable extrapolation, but only while the signed target remains

realizable. The extrapolation identity is geometric: moving beyond m_+ does not by itself guarantee greater task utility. At the critical regime $p = q$, attraction and repulsion cancel only in total mass, leaving a nonzero moment difference that drives linear drift. Under negative dominance, the objective becomes locally convex rather than concave around any stationary point, making that point repelling. Formally, the local Jacobian for $p > q$ is $-(p - q)\nabla^2\psi(\eta^*)$, while the discrete step requires $\rho(I + \alpha J_F) < 1$. Appendix A.3 gives the existence, boundary, and exact step-size arguments. The theorem classifies the aggregate regimes; the next result derives their policy-family manifestations.

Theorem 4.3 shows that negative dominance admits no finite stable equilibrium, but it does not determine how runaway appears in a particular policy family. Gaussian and categorical policies share the aggregate instability while differing sharply in gradient amplitude and boundary behavior. The distinction is structural.

Theorem 4.4 (Unbounded Gaussian and bounded-score categorical runaway). *Assume $p < q$ and let $r = q - p > 0$. For a fixed-covariance Gaussian policy, define $\mu^\dagger = (q\mu_- - p\mu_+)/r$. The aggregate mean field is*

$$\mathbf{F}_\mu(\mu) = r\Sigma^{-1}(\mu - \mu^\dagger). \quad (14)$$

Unless initialized exactly at $\mu^\dagger$, continuous and discrete trajectories satisfy $\|\mu\|_2 \rightarrow \infty$, and for every fixed historical action a , $\|\nabla_\mu \log \pi_\mu(a)\|_2 \rightarrow \infty$. For a categorical policy with gauge-fixed logits, every nonstationary aggregate trajectory leaves each compact subset of logit space and has a subsequence approaching the simplex boundary, while

$$\|\nabla_z \log \pi_z(a)\|_2^2 = \|e_a - \pi_z\|_2^2 \leq 2. \quad (15)$$

In the Gaussian case, $\mu^\dagger$ is a repelling point and the positive-definite matrix $r\Sigma^{-1}$ expands every nonzero displacement. Increasing separation from any fixed action then converts directly into gradient-amplitude runaway. In the categorical case, fixing the additive-logit gauge reveals strictly convex ascent away from any finite stationary point. The logits become unbounded and probabilities approach the simplex boundary, but each direct-logit score remains less than or equal to $\sqrt{2}$. Categorical degeneration is therefore support suppression, not Gaussian-style gradient explosion. This bounded-score distinction also prevents us from interpreting a probability-boundary event as evidence of floating-point overflow or an unbounded Euclidean token gradient. A learned Gaussian covariance can further amplify standardized distance when support contracts, but covariance learning is not required for the fixed-covariance mean result. Neither theorem branch alone establishes task-performance collapse or NaN/Inf numerical failure; those events require separate empirical tests. Appendix A.4 provides both derivations, with additional Gaussian and categorical details in

Appendices B and C. These distinctions lead directly to the predictions below.

The preceding results imply a falsifiable phase sequence rather than a generic prediction that negative updates are bad. Positive-only training should reach a finite platform determined by the observed positive target. Under Theorem 4.3, moderate negative influence should produce stable extrapolation to a second finite platform when the signed moment remains feasible. Increasing negative mass or moving its moment outward should bring the policy toward a support, variance, or probability boundary; after equilibrium loss, Theorems 4.2–4.4 predict persistent drift or a non-vanishing terminal residual rather than a genuine stationary state. The causal prediction is selective: if the far-field component transmits the instability, removing or capping the far path should rescue a finite terminal regime, whereas removing only near-field negatives should not. Global scaling and budget-matched transfers then test whether total magnitude or learner-relative location is the operative mediator. Proposition 5.1 predicts that the DRPO envelope removes any finite-order far-field tail, but it does not pre-rank all taper shapes at finite distance. Every experiment must separately report task-performance collapse, support or variance-boundary events, and NaN/Inf numerical failure. C-U1 and D-U1 provide controlled ground-truth tests of these predictions; Hopper and Countdown assess external validity under learned critics and shared sequence parameters. The controlled environments do not substitute for the external tasks, and finite-step external pilots do not establish terminal method rankings.

5. Distributionally Robust Policy Optimization

DRPO applies a policy-relative envelope to the negative component of the signed actor field. Let $r_\theta(s, a) \geq 0$ denote the registered remoteness measure for the policy family and experiment. We define

$$w_\lambda^-(s, a) = \exp[-\lambda r_\theta(s, a)], \quad \lambda \geq 0, \quad (16)$$

and optimize

$$\mathbf{F}_{\text{DRPO}}(\theta) = \mathbb{E}_\nu[A^+ \nabla \log \pi_\theta - A^- w_\lambda^- \nabla \log \pi_\theta]. \quad (17)$$

Positive updates are unchanged, nearby negative actions retain substantial influence, and increasingly remote negative actions are attenuated. Sample quality and learner-relative remoteness remain distinct axes: the matched experiments vary remoteness without changing badness, while Appendix H states the separate Optimistic-DRO quality-selection result. The main DRPO update does not equate the two controls.

In the exponential-family analysis, the uncontrolled negative moment is

$$q\mathbf{m}_- = \mathbb{E}[A^- T(a)]. \quad (18)$$

DRPO replaces it by

$$q_\lambda \mathbf{m}_{-, \lambda} = \mathbb{E}[A^- e^{-\lambda r_\theta(s, a)} T(a)], \quad (19)$$

which induces the controlled target

$$\mathbf{m}_\lambda^* = \frac{p \mathbf{m}_+ - q_\lambda \mathbf{m}_{-, \lambda}}{p - q_\lambda}. \quad (20)$$

The method therefore modifies the same term that first creates stable extrapolation and later drives the signed target toward the boundary. Small r_θ preserves substantial negative influence; large r_θ is selectively attenuated.

Proposition 5.1 (Vanishing weighted far-field gradient). *Suppose the policy score satisfies $\|\nabla_\theta \log \pi_\theta(a \mid s)\| \leq C(1 + r_\theta(s, a))^k$ for finite C, k . For any $\lambda > 0$,*

$$e^{-\lambda r_\theta(s, a)} \|\nabla_\theta \log \pi_\theta(a \mid s)\| \rightarrow 0 \quad \text{as } r_\theta(s, a) \rightarrow \infty. \quad (21)$$

The exponential form is therefore a gradient-tail envelope derived from the far-field score-growth bound. Directional utility is measured separately in the controlled experiments.

We compare DRPO against an uncontrolled signed baseline, Positive-only learning, non-selective global scaling, registered linear or reciprocal tapers, and a hard threshold on the same remoteness coordinate. A quality-based hard filter is reported separately because it controls a different axis. Where the protocol calls for budget matching, methods receive the same pre-optimizer raw negative-gradient norm and are evaluated with paired seeds. No method ranking is assumed in advance; best and terminal checkpoints, local-negative retention, far-field retention, task performance, boundary events, and numerical failures are all reported.

6. Experiments

We use two primary controlled environments and two external families. C-U1 is a nonlinear state-conditioned Gaussian environment with six-dimensional numerical contexts, two-dimensional actions, a known state-dependent optimum, and independently sampled train and test contexts from the same distribution. It supports matched quality–distance probes, near/far intervention, and mean–support terminal audits. D-U1 is a shared-network categorical environment with unordered semantic actions and a known reward structure; it supports matched quality–rarity probes, common/rare intervention, and probability-boundary audits. We report C-U1 using the precise term held-out-context or unseen-state generalization. Hopper/D4RL adds a learned critic and long-horizon continuous control, while Countdown adds shared Transformer parameters and sequence-level verification. External tasks establish relevance; controlled environments establish source and causality.

Hopper/D4RL. We train the registered learned critic, freeze its selected checkpoint, and audit negative actor gradients by current-policy distance. The formal report will include far/near negative-gradient ratios, positive/negative imbalance, policy drift, rollout return, and terminal classification. **TBD:** insert only terminal-audited formal results from the registered E7 package.

Countdown. We bin negative tokens or completions by current-policy surprisal and track score magnitude, shared-parameter gradient influence, target-probability change, verifier success, validity, and pass@ k . **TBD:** insert only the registered formal result; focused pilots remain provenance rather than final evidence.

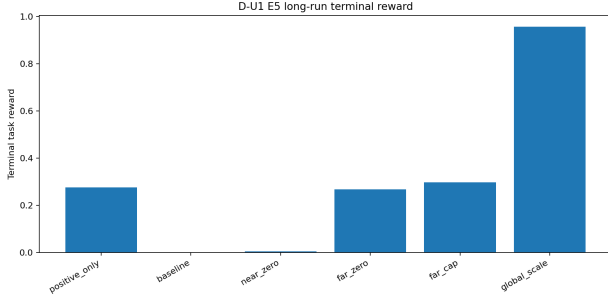
The source-isolation protocol holds sample badness fixed. In C-U1, negative actions share the same state, reward, advantage magnitude, quality coordinate, count, coefficient, and policy parameters while varying only learner-relative radius. We report output-space score norms, full-parameter per-sample gradients, aggregate gradients, and directional coherence. In D-U1, common and rare actions are matched in semantic role, reward, negative advantage, count, and base coefficient while initial probability or current surprisal changes. A far/near or rare/common gap under these controls identifies policy remoteness as an independent amplifier; it does not imply that distance is the only factor in real data.

We next intervene on the location of negative influence. The continuous comparison includes the uncontrolled signed baseline, Near-zero, Far-zero, Far-cap, a non-selective global control matched to the retained raw negative-gradient budget, and a registered budget-transfer control. The categorical analogue suppresses common or rare negatives under matched counts and budgets. We record the onset order of remote influence, parameter drift, support or probability boundaries, and task performance. A selective far/rare rescue together with an ineffective near/common removal supports the remote component as a causal transmission pathway; task-performance collapse, boundary events, and NaN/Inf remain distinct labels.

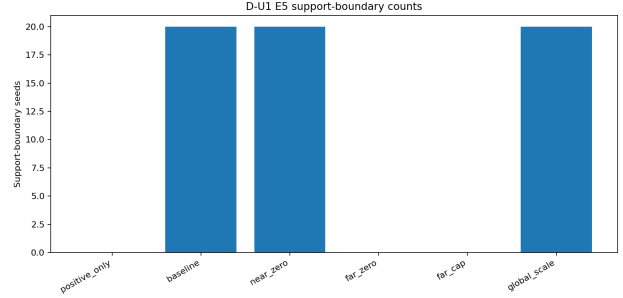
We sweep the effective negative strength from zero through controlled repulsion and into the unstable regime. The primary test is not a single endpoint score but the phase sequence predicted by Theorem 4.3: Positive-only, stable extrapolation, boundary approach, and persistent drift or loss of finite equilibrium. We measure a family-specific empirical proxy for the aggregate negative term in (19), together with its distance or rarity bins, direction, and relationship to equilibrium displacement. DRPO is compared with Positive-only, global scaling, and registered distance controls under paired seeds and explicit raw-gradient budgets. Results are reported at both selected and terminal

Table 1. Evidence roles are separated by design. Controlled environments identify source and causality; external environments establish relevance and task effect.

Environment	Primary responsibility	Complementary boundary
C-U1	Continuous matched source isolation, near/far causality, and phase transition	External relevance is established in Hopper/D4RL; categorical boundaries are tested in D-U1
D-U1	Categorical matched rarity isolation, common/rare causality, and probability boundary	Gaussian score-amplitude dynamics are tested in C-U1
Hopper/D4RL	External continuous-control occurrence and task effect with a learned critic	Exact badness–distance isolation is supplied by C-U1
Countdown	External shared-parameter sequence occurrence and task effect	Controlled common/rare causality is supplied by D-U1



(a) Terminal reward.



(b) Support-boundary events.

Figure 1. D-U1 long-run causal results. Baseline and Near-zero preserve the rare/far negative pathway and reach the support boundary in 20/20 seeds; Far-zero and Far-cap remain bounded in 20/20 seeds. Global scaling can preserve task reward while still reaching the support boundary, illustrating why task, boundary, and numerical events must be reported separately.

Table 2. D-U1 E5 long-run summary.

Method	Reward	Task collapse	Boundary
baseline	0.001	20/20	20/20
near-zero	0.006	20/20	20/20
far-zero	0.267	0/20	0/20
far-cap	0.297	0/20	0/20
global-scale	0.957	0/20	20/20
positive-only	0.275	0/20	0/20

Table 3. C-U1 intervention summary (20 seeds).

Method	Reward	Task event	Boundary event
baseline	0.603	0.00	1.00
near-zero	0.602	0.00	1.00
far-zero	0.653	0.00	0.00
far-cap	0.653	0.00	0.00
global-scale	0.643	0.00	0.00

checkpoints. **TBD:** populate the final table only from the current terminal-audited repository artifacts and preserve each result’s formal status.

The final evaluation returns to external tasks. On D4RL locomotion, methods share dataset, critic protocol, initialization, seeds, evaluation horizon, and selection rule; we report normalized return, uncertainty, best and terminal checkpoints, and distance-binned negative influence. On Countdown,

methods share the SFT initialization, rollout or replay bank, verifier, seeds, and checkpoint-selection protocol; we report greedy success, pass@ k , validity, terminal degradation, and rare-negative diagnostics. **TBD:** the main external tables remain unfilled until the corresponding registered formal experiments are terminal-audited and delivered.

7. Implications and Conclusion

Negative feedback is a policy-improvement resource with a dynamical failure mode. It provides boundary information and can shift a policy beyond the Positive-only target. The failure arises when historical actions remain active after becoming remote from the current learner, so their optimization influence no longer tracks their local relevance. The transferable design principle is therefore to control learner-relative far-field negative influence rather than delete negative feedback as a class.

The continuous and categorical analyses share an aggregate structure but not an identical local law. Gaussian policies can amplify mean scores with standardized distance and couple repulsion to support contraction. Categorical direct-logit scores remain bounded, yet repeated negative updates can persistently suppress probability. In both cases, aggregate positive–negative competition determines whether the policy remains at an interior equilibrium or moves toward a

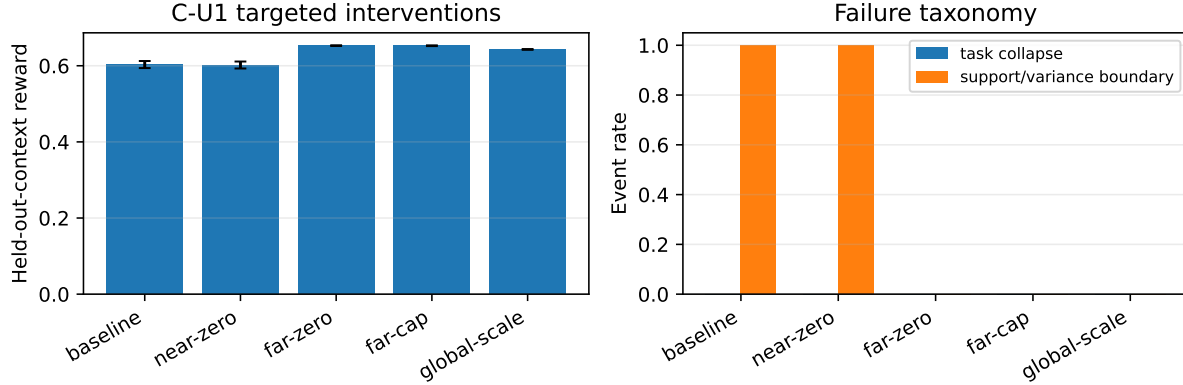


Figure 2. C-U1 causal intervention with learnable variance. The left panel reports registered held-out-context reward with confidence intervals; the right panel separates task-performance collapse from support or variance-boundary events. Near-zero tracks the uncontrolled baseline, while Far-zero and Far-cap remove the task-collapse pathway without introducing NaN/Inf failures.

Table 4. Controlled phase and method results. Entries remain TBD until the corresponding registered artifacts and terminal audits are frozen for the manuscript. Task collapse, boundary events, and numerical failure are reported separately.

Method	Held-out-context reward	Near retention	Far retention	Task collapse	Boundary event	NaN/Inf
Positive-only	TBD	–	0	TBD	TBD	TBD
Uncontrolled signed	TBD	TBD	TBD	TBD	TBD	TBD
Global matched	TBD	TBD	TBD	TBD	TBD	TBD
Hard distance control	TBD	TBD	TBD	TBD	TBD	TBD
DRPO exponential	TBD	TBD	TBD	TBD	TBD	TBD

feasibility boundary.

Negative feedback can create stable policy improvement beyond Positive-only learning. By separating badness from policy remoteness, we identify how historical negative actions become far-field, dominate the aggregate update, and move the policy toward a feasibility boundary where a finite equilibrium can disappear. DRPO controls this same negative term, preserving useful local repulsion while suppressing the destructive far-field tail.

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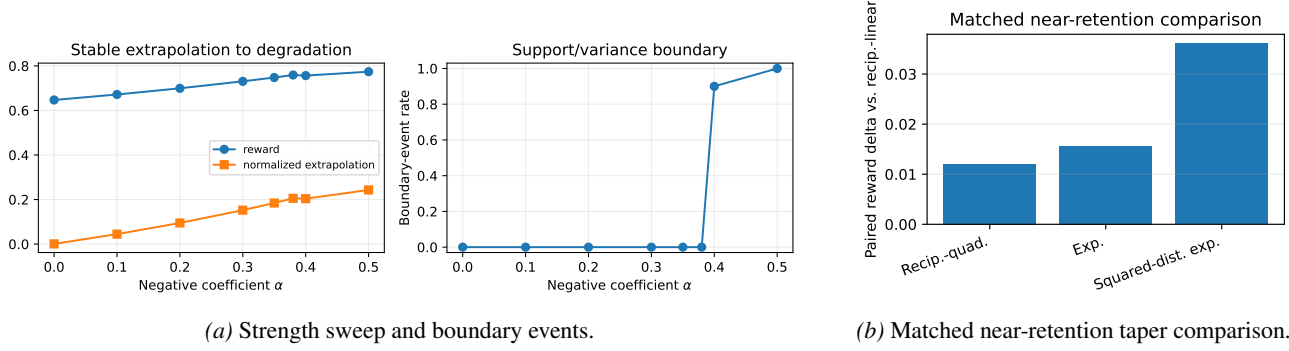


Figure 3. C-U1 phase and control evidence. The strength sweep maps the transition from the Positive-only reference through stable extrapolation toward boundary events. Under the frozen near-retention protocol, all three faster-decaying tapers improve finite-horizon held-out-context reward on 20/20 paired seeds relative to reciprocal-linear; this result remains finite-step validated rather than a universal steady-state ranking.

Table 5. External task closure. Best and terminal results are both required; no values are inserted before formal terminal-audited completion.

Family	Dataset/model	Uncontrolled	Positive-only	Global	DRPO	Terminal failure rate
D4RL	Hopper / locomotion suite	TBD	TBD	TBD	TBD	TBD
Countdown	Main model scale	TBD	TBD	TBD	TBD	TBD

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A. Proofs for Repulsive Dynamics

For a regular minimal exponential family, $\nabla\psi$ maps the natural-parameter space diffeomorphically onto the interior of the mean-parameter space. Setting (11) to zero with $p > q$ gives (12). If $\mathbf{m}^*$ is interior, uniqueness follows from strict convexity of ψ . The displacement identity follows by subtracting $\mathbf{m}_+$ from (12). If $\mathbf{m}^*$ approaches the boundary, no finite natural parameter realizes it; if it lies outside the closure, no policy in the family realizes the target. Differentiating (11) gives $J_F = -(p - q)\nabla^2\psi$, which is negative definite at an interior point because the Hessian is the positive-definite covariance of the sufficient statistic. The discrete update is locally stable for step sizes whose transition matrix has spectral radius below one.

A.1. Proof of Proposition 4.1

Proof. Let $\delta = a - \mu$. For a Gaussian with fixed covariance,

$$\begin{aligned} D_\mu(a) - C_\Sigma &= \frac{1}{2}\delta^\top \Sigma^{-1}\delta, \\ R_\mu(a) &= \|\Sigma^{-1}\delta\|_2^2 = \delta^\top \Sigma^{-2}\delta. \end{aligned} \quad (22)$$

With $x = \Sigma^{-1/2}\delta$, these become $D_\mu - C_\Sigma = \|x\|_2^2/2$ and $R_\mu = x^\top \Sigma^{-1}x$. The Rayleigh quotient therefore gives

$$\begin{aligned} R_\mu(a) &\geq 2\lambda_{\min}(\Sigma^{-1})(D_\mu - C_\Sigma), \\ R_\mu(a) &\leq 2\lambda_{\max}(\Sigma^{-1})(D_\mu - C_\Sigma). \end{aligned} \quad (23)$$

If $\delta = rv$ for a fixed direction, eliminating r^2 gives the exact directional coefficient; when $\Sigma = \sigma^2 I$, it reduces to $2/\sigma^2$. For a categorical policy, $\nabla_z \log \pi_z(a) = e_a - \pi_z$. Writing $p_a = e^{-D_z(a)}$ yields $|\partial_{z_a} \log \pi_z(a)| = 1 - p_a = 1 - e^{-D_z(a)}$. Moreover,

$$\|e_a - \pi_z\|_2^2 = (1 - p_a)^2 + \sum_{j \neq a} \pi_z(j)^2 \leq 2(1 - p_a)^2 \leq 2. \quad (24)$$

Thus the selected-logit response increases with surprisal but remains bounded. $\square$

A.2. Proof of Theorem 4.2

Proof. Write $f(u) = D(u)$, $g_t = \nabla f(u_t)$, and $R_t = \|g_t\|_2^2$. For a negative coefficient $\hat{A} = -c$ with $c > 0$, the update is $u_{t+1} = u_t + \alpha g_t$, where $\alpha = \eta c$. Convexity gives

$$f(u_{t+1}) \geq f(u_t) + g_t^\top (u_{t+1} - u_t) = f(u_t) + \alpha \|g_t\|_2^2. \quad (25)$$

The gradient of a differentiable convex function is monotone, hence $(g_{t+1} - g_t)^\top (u_{t+1} - u_t) \geq 0$. Substituting $u_{t+1} - u_t = \alpha g_t$ gives $g_{t+1}^\top g_t \geq \|g_t\|_2^2$; Cauchy-Schwarz then implies $\|g_{t+1}\|_2 \geq \|g_t\|_2$. For a positive coefficient $\hat{A} = c > 0$, the update is $u_{t+1} = u_t - \alpha g_t$. The descent lemma yields

$$f(u_{t+1}) \leq f(u_t) - \alpha(1 - L\alpha/2)\|g_t\|_2^2 \leq f(u_t) - \frac{\alpha}{2}R_t \quad (26)$$

when $\alpha \leq 1/L$. For a convex function with L -Lipschitz gradient, cocoercivity gives $(g_t - g_{t+1})^\top (u_t - u_{t+1}) \geq L^{-1}\|g_t - g_{t+1}\|_2^2$. Using $u_t - u_{t+1} = \alpha g_t$ and expanding $\|g_{t+1}\|_2^2$ proves $R_{t+1} \leq R_t$ for $\alpha L \leq 1$. $\square$

A.3. Detailed derivation for Theorem 4.3

Proof. The signed population objective is

$$L(\eta) = (p\mathbf{m}_+ - q\mathbf{m}_-)^\top \eta - (p - q)\psi(\eta) + C, \quad (27)$$

so $F(\eta) = \nabla L(\eta) = p\mathbf{m}_+ - q\mathbf{m}_- - (p - q)\nabla\psi(\eta)$ and $\nabla^2 L(\eta) = -(p - q)\nabla^2\psi(\eta)$. For $p > q$, $F(\eta) = 0$ is equivalent to

$$\nabla\psi(\eta) = \mathbf{m}^* = \frac{p\mathbf{m}_+ - q\mathbf{m}_-}{p - q}. \quad (28)$$

A regular minimal exponential family has strictly convex log-partition ψ , so $\nabla\psi$ is one-to-one and maps the natural-parameter space onto the interior mean-parameter set. An interior $\mathbf{m}^*$ therefore has a unique finite preimage η^* . Subtracting $\mathbf{m}_+$ yields

$$\mathbf{m}^* - \mathbf{m}_+ = \frac{q}{p - q}(\mathbf{m}_+ - \mathbf{m}_-). \quad (29)$$

The field Jacobian is $J_F(\eta^*) = -(p - q)\nabla^2\psi(\eta^*) \prec 0$, proving continuous-time local asymptotic stability. The discrete Jacobian is $I - \alpha(p - q)\nabla^2\psi(\eta^*)$, whose spectral radius is below one whenever

$$0 < \alpha < \frac{2}{(p - q)\lambda_{\max}(\nabla^2\psi(\eta^*))}. \quad (30)$$

If $\mathbf{m}^*$ approaches the feasible boundary, continuity and invertibility imply that its natural parameter leaves every compact subset; outside the feasible set no finite solution exists. When $p = q$, the field is the constant $p(\mathbf{m}_+ - \mathbf{m}_-)$. It produces linear drift if the moments differ and a non-isolated stationary continuum otherwise. When $p < q$, $\nabla^2 L = (q - p)\nabla^2\psi \succ 0$; any finite stationary point has a positive-definite continuous Jacobian and a discrete Jacobian with eigenvalues strictly larger than one, so it is unstable. $\square$

A.4. Proof of Theorem 4.4

Proof. Let $r = q - p > 0$. For the fixed-covariance Gaussian,

$$F_\mu(\mu) = \Sigma^{-1}[p\mu_+ - q\mu_- - (p - q)\mu] = r\Sigma^{-1}(\mu - \mu^\dagger). \quad (31)$$

With $\delta = \mu - \mu^\dagger$, continuous time gives $\dot{\delta} = r\Sigma^{-1}\delta$ and hence $\delta(t) = \exp(r\Sigma^{-1}t)\delta(0)$. Every nonzero displacement grows at least as $\exp(r\lambda_{\min}(\Sigma^{-1})t)$. The discrete recurrence is $\delta_{t+1} = (I + \alpha r\Sigma^{-1})\delta_t$, whose eigenvalues exceed one for every $\alpha > 0$. Thus every nonstationary trajectory diverges. For fixed a , $\|\nabla_\mu \log \pi_\mu(a)\|_2 = \|\Sigma^{-1}(a - \mu)\|_2 \rightarrow \infty$.

For the categorical policy, fix the gauge $\mathbf{1}^\top z = 0$ and write the signed objective as

$$L(z) = b^\top z + r \log \sum_j e^{z_j} + C. \quad (32)$$

Its Hessian on the gauge-fixed subspace is $r(\text{Diag}(\pi_z) - \pi_z \pi_z^\top) \succ 0$, so L is strictly convex there. Along continuous gradient ascent, $dL/dt = \|\nabla L\|_2^2$; along the discrete update, convexity gives $L(z_{t+1}) \geq L(z_t) + \alpha \|\nabla L(z_t)\|_2^2$. A bounded nonstationary trajectory would therefore have an accumulation point with zero gradient. If a finite stationary point exists, strict convexity makes it unique and the squared distance from it increases under ascent; if none exists, zero-gradient accumulation is impossible. Hence every nonstationary trajectory leaves each compact set. On the gauge-fixed subspace, probabilities bounded away from zero imply bounded logits, so an unbounded trajectory has a subsequence approaching the simplex boundary. Finally, $\|e_a - \pi_z\|_2^2 \leq 2(1 - \pi_z(a))^2 \leq 2$, proving bounded per-sample scores. $\square$

A.5. Proof of Proposition 5.1

Proof. Assume the unweighted score satisfies $\|\nabla_\theta \log \pi_\theta(a \mid s)\| \leq C(1 + r_\theta)^k$ for finite k , and let $w_\lambda(r) = \exp(-\lambda r^2)$. Then

$$\|w_\lambda(r) A^\top \nabla_\theta \log \pi_\theta\| \leq |A^\top| C(1 + r)^k e^{-\lambda r^2}. \quad (33)$$

For every finite k and $\lambda > 0$, the Gaussian tail dominates the polynomial factor, so the right-hand side converges to zero as $r \rightarrow \infty$. This proves vanishing weighted far-field influence without assuming that the sample’s utility decays exponentially. $\square$

B. Gaussian Mean–Variance Derivations

For a scalar Gaussian with $\xi = \log \sigma$ and $z = (a - \mu)/\sigma$,

$$\partial_\mu \log \pi = \frac{a - \mu}{\sigma^2}, \quad \partial_\xi \log \pi = z^2 - 1. \quad (34)$$

A negative advantage reverses both score directions. It always pushes the mean away from the action, but its effect on scale depends on $|z|$: a near negative action with $|z| < 1$ increases σ , whereas a far negative action with $|z| > 1$ decreases σ . Positive updates have the opposite signs. A full Gaussian terminal state must satisfy both mean and variance equations; a stationary mean with continuing covariance movement is not a complete policy equilibrium.

B.1. Population mean–variance fixed point

With $\xi = \log \sigma$, positive mass p , effective negative mass q , conditional moments (m_+, v_+) and (m_-, v_-) . Define

$M_\pm(\mu) = v_\pm + (\mu - m_\pm)^2$. Then

$$\dot{\mu} = \frac{p(m_+ - \mu) - q(m_- - \mu)}{\sigma^2}, \quad (35)$$

$$\dot{\xi} = \frac{pM_+(\mu) - qM_-(\mu)}{\sigma^2} - (p - q). \quad (36)$$

For $p > q$, the mean candidate is $\mu^* = (pm_+ - qm_-)/(p - q)$. Substitution into the scale equation gives

$$(\sigma^2)^* = \frac{pM_+(\mu^*) - qM_-(\mu^*)}{p - q}, \quad (37)$$

A finite joint equilibrium therefore requires both $p > q$ and a positive numerator. This second condition is stricter in the far field because M_- contains squared displacement.

C. Categorical Boundary Dynamics

For softmax probabilities p_j , the score for selected action y is $e_y - p$, whose Euclidean norm is bounded. Under a negative update with fixed local coefficient, the selected logit decreases relative to competing logits, so $z_y - z_j$ falls approximately linearly while p_y can decay exponentially. The failure mode is therefore persistent support suppression toward the simplex boundary. In a shared network, the parameter-space gradient additionally contains the network Jacobian and cross-action interference, which are measured rather than inferred from the direct-logit bound alone.

C.1. Repeated negative updates and log-odds

For softmax logits z , $\nabla_z \log p_y = e_y - p$. Under a fixed negative coefficient $c > 0$ and small step size h , the selected logit changes by $-hc(1 - p_y)$ while competitor j changes by $+hcp_j$. Hence

$$(z_y - z_j)_{t+1} - (z_y - z_j)_t = -hc(1 - p_y + p_j), \quad (38)$$

which remains negative as $p_y \rightarrow 0$. The Euclidean score is bounded, but the non-vanishing log-odds decrement yields persistent support suppression and can drive p_y exponentially toward the simplex boundary.

D. Controlled Environments

C-U1 samples numerical contexts $s \sim \mathcal{N}(0, I_6)$ independently for training and testing. A fixed generator maps each state to a positive action, hidden optimal action, negative quality coordinate, and reward surface in $\mathbb{R}^2$. Source-isolation probes replicate reward and advantage across radii while retaining the same state and quality coordinate. Near/far membership for causal interventions is computed relative to the current policy according to the registered metric. D-U1 uses a shared state-conditioned categorical network over unordered semantic actions; quality/semantics

and rarity are factorized so that common/rare probes can be matched in reward, advantage, count, and coefficient. Complete constants, seeds, and invariants are taken from the registered experiment configurations.

E. Experimental Protocols and Terminal Audits

Formal protocols specify maximum horizons, evaluation cadence, paired development and held-out seeds, and a terminal audit before execution. A fixed horizon is not itself called convergence. Depending on the registered experiment, terminal classification uses state slopes, update residuals, boundary checks, and a continuation horizon. Budget-matched comparisons use the pre-optimizer raw negative-gradient ℓ_2 norm unless another coordinate is explicitly frozen. Adam parameter-update norms are recorded separately. Best-validation and terminal checkpoints are both reported, and task-performance collapse, support or variance boundaries, and NaN/Inf failures remain distinct fields.

F. Additional Results and Failure Taxonomy

This appendix will contain per-seed results, full training trajectories, confidence intervals, sensitivity analyses, and terminal classifications for every main comparison. Failed or inconclusive formal runs are indexed rather than silently removed. **TBD:** populate these materials from the durable formal packages after repository closure; smoke tests, static checks, and focused pilots are not promoted to formal evidence.

G. Implementation and Reproducibility

The repository records experiment IDs, configurations, seeds, code entry points, result manifests, raw curves, logs, terminal audits, checksums, and source commits. Formal runs progress through registered, running, raw-complete, terminal-audited, packaged, delivered, and applied states. The paper source is generated from the stable-ID manuscript graph, compiled in the imported ICML/Overleaf project, and packaged with a reproducible build command. Exact repository paths and commit hashes will be inserted at submission freeze.

H. Optimistic Distributional Selection

Let P denote the empirical data distribution and let $R(s, a)$ be the registered quality signal. The Optimistic-DRO selec-

tion problem is

$$\begin{aligned} \max_Q \quad & \mathbb{E}_Q[R(s, a)] \\ \text{s.t.} \quad & Q \ll P, \quad 0 \leq \frac{dQ}{dP} \leq \frac{1}{\kappa}, \\ & \int dQ = 1. \end{aligned} \quad (39)$$

Its optimum allocates the maximum admissible density to the highest-quality region of P , with boundary mass only when needed to satisfy normalization. For a continuous reward distribution this is the top- κ hard-selection solution. This result establishes a principled quality-selection operator inside the current DRPO manuscript. It does not identify policy remoteness with low quality and does not derive the exponential envelope in Eq. (16). Quality selection controls which data are emphasized; remoteness tapering controls how a negative sample’s influence changes as the learner moves. The source-isolation experiments keep quality fixed precisely so that the second effect can be identified independently.

I. Additional Theoretical and Reporting Clarifications

The manuscript uses the following technical conventions throughout. Far-field negative Gaussian updates combine mean repulsion with location-dependent scale dynamics and commonly support contraction; they do not universally expand both mean and variance. The expected Fisher matrix describes policy geometry but does not by itself determine stability of a fixed off-policy signed field; local stability is evaluated with the Jacobian of the actual aggregate field. C-U1 test states are independently sampled from the same state distribution and are described as held-out-context or unseen-state generalization, not OOD generalization. Task-performance collapse, support or variance boundaries, and NaN/Inf numerical failure are reported separately. Finally, finite-horizon evidence is not relabeled as convergence or long-run validation without the registered terminal audit.